Polynomial Inequalities

We're used to solving inequalities that are linear $(\text{i.e. } 2x-1>0)$, but today we are solving polynomial inequalities of higher degree than 1.

To solve:

1. Find all the real zeros of the polynomial (not imaginary) ** zeros = "critical points"
2. Plot the zeros on a number line (remember open or closed circles)
3. Pick one x-value in each test interval (determined by the critical points) and evaluate the sign at that interval.
4. Write your answer in interval notation ( ) = open circles & +/- infinity, [ ] = closed circles

Example 1: Solve the following polynomial inequalities:

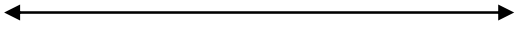
A) $x^2-5x-6<0$



B) $2x^3-x^2-32x\geq -16$



C) $(x-2)^2(x+3)>0$



D) $x^4 - 4x^2 + 3 > 0$



E) $x^2 + 6x + 9 > 0$



F) $x^2 + 4x + 4 \leq 0$



G) $x^2 + 5x + 7 < 0$



H) $x^2 - 6x + 9 \geq 0$

